ENGR 3430: Miniproject 3

layout due October 16, 2026
demo due November 13, 2026

In this miniproject, you will gain more experience doing surface-mount printed circuit board (PCB) layout and assembly with a more complex circuit. You will be designing a Raspberry-Pi-Pico-2-like microcontroller development board, which is shown in the attached schematic, based on a KiCad project that you can download from the "Files" section of the course Canvas site. Other than a maximum size constraint of about 4 in × 4 in imposed by the board fabricator (PCBWay), there are no externally imposed board size considerations. For the tactile switches, resistors, and most capacitors, we will have two size options available for you use to assemble your boards. The list of available components for MP3 will be provided as a spreadsheet that you can download from the course Canvas site. We will submit your boards for fabrication through PCBWay. When they come back, you will assemble your board, program the microcontroller to blink the user LED at 1 Hz and submit a demo video of the code running on your board.

Requirements. Your design must meet the following requirements:

1. It must be a two-layer board, no larger than 3.94 in × 3.94 in (i.e., 100 mm × 100 mm) with all components on the top side.
2. The minimum allowable trace width and spacing is 6 mils.
3. The minimum allowable via size is 24 mils with a 12-mil drill hole.

Deliverables. Your final PCB layout is due by the start of class on Friday, October 16. You will need to submit copies of all of your KiCad files. Assuming there are no problems with your design, we will generate the Gerber files and submit them to PCBWay for fabrication. The boards should be back from fab by the end of October. Once they return, you will assemble your board, get an appropriate firmware development toolchain up and running to write code, and demonstrate the blinking LED running on your board by no later than the start of class on Friday, November 13.